Amazon Inventory Reconciliation Using AI: Counting Items in Bin Images

UE24CS352A Machine Learning, Mini-Project
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Code: github.com/tejasshetty23/amazon-bin-reconciliation (private, shared with faculty)

1. Problem statement

Items in Amazon warehouse bins are sometimes misplaced, so the recorded count no longer matches the bin. From a single photo of a bin, we predict how many items it holds (0-5) and flag bins whose recorded count looks wrong. We reproduce and extend the Stanford CS229 study by Rodriguez Bertorello et al.

2. Dataset

Amazon Bin Image Dataset (AWS Open Data, about 535K photos; label = EXPECTED_QUANTITY). We sampled 20,000 bins with 0-5 items (6 classes, 16.7% random baseline), resized them to 256x256 and split them 70/20/10, stratified. Empty bins are rare (3%), and many items are hidden behind elastic tape.

3. Approach

- **Classical baselines:** raw pixels + PCA, HOG (also on edge maps), Laplacian-of-Gaussian blob statistics and colour histograms, fed to logistic regression, a decision tree and linear / RBF SVMs tuned by cross-validated grid search.
- **CNNs (transfer learning):** ImageNet-pretrained ResNet18 / ResNet34 with a new 6-way head. One epoch trains only the head, then the whole network is fine-tuned with layer-wise learning rates. We compared SGD + step decay, SGD with warm restarts (SGDR), and AdamW + one-cycle with class weights. Mild augmentation only (flips, small shifts, colour jitter), early stopping, and flip test-time augmentation.
- **Reconciliation (our extension):** flag a bin when the model's probability for the *recorded* count is below a threshold; evaluated on simulated records with 50% errors. We also report RMSE, off-by-one accuracy and macro-F1.

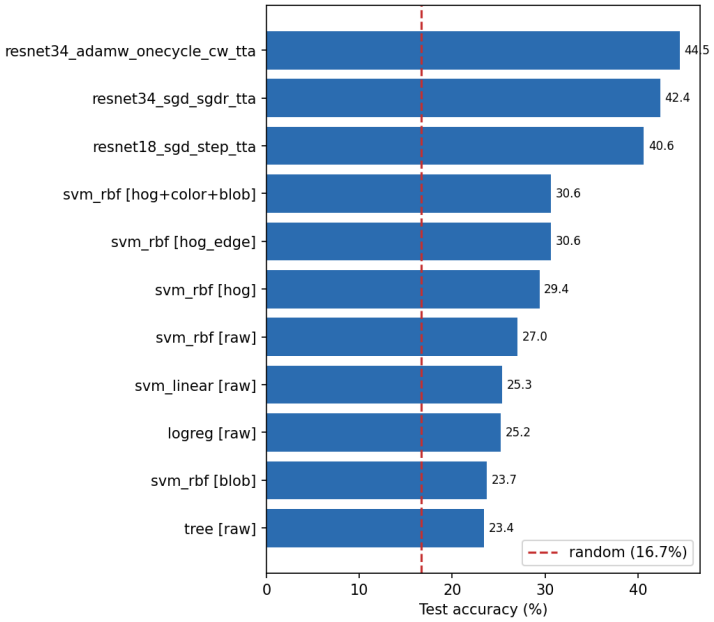
4. Implementation

Python, scikit-learn, scikit-image, PyTorch. One-command pipeline (download, split, features, training, evaluation, summary), trained on a Colab T4 GPU, with smoke tests on synthetic data. A Gradio demo takes a bin photo and its recorded count and returns the predicted count, a MATCH / FLAG FOR AUDIT verdict and a Grad-CAM heat-map.

5. Results (held-out test split)

Model (test split, flip TTA)	Acc.	Off-by-one	RMSE	Macro-F1	Recon. AUC
resnet34_adamw_onecycle_cw	**44.5%**	**83.8%**	**1.109**	**0.534**	**0.727**
resnet34_sgd_sgdr	42.4%	83.9%	1.092	0.505	0.705
resnet18_sgd_step	40.6%	84.0%	1.099	0.497	0.699
svm_rbf [hog+color+blob]	30.6%	73.6%	1.353	0.390	-
svm_rbf [hog_edge]	30.6%	67.3%	1.549	0.358	-
svm_rbf [hog]	29.4%	71.9%	1.408	0.354	-
Paper: ResNet34 + Adam, 324K imgs (reported)	56.2%	-	0.900	-	-
Random guess (6 classes)	16.7%	-	-	-	-

Paper row is a reported number on a different sample, for context only. Recon. AUC = how well P(count = recorded) separates wrong records from correct ones.



6. Conclusions

- **Learned features win:** best CNN 44.5% vs best classical 30.6% (2.7x random). HOG and blob features are confused by the tape and clutter.
- **Errors are mostly small:** 83.8% of predictions are within one item. Empty bins are easiest (98.2%); middle counts are hardest, since they have neighbours on both sides.
- **Reconciliation:** mismatch-detection AUC 0.73. At threshold 0.5 the model over-flags (recall 89.9%, precision 56.8%), so it is best used to rank bins for audit, not as an automatic check.
- **Data is the bottleneck:** we trained on 14K images vs the paper's 105K-260K; training accuracy kept rising while validation plateaued. The CNN variants differ by only a few points, close to test-set noise.

7. Individual contributions

Tejas Shetty: data pipeline, feature engineering, classical baselines, metrics and reconciliation evaluation, write-up. **Tanishk Gowda:** CNN training pipeline, GPU experiments, error analysis, demo, slides. **Both:** experiment design, analysis, presentation.

References: Rodriguez Bertorello, Sripada, Dendumrongsup, *Amazon Inventory Reconciliation Using AI* (Stanford CS229); Amazon Bin Image Dataset (AWS Open Data); He et al., ResNet (CVPR 2016); Loshchilov & Hutter, SGDR (ICLR 2017); Smith, Cyclical LR (WACV 2017); Selvaraju et al., Grad-CAM (ICCV 2017); Dalal & Triggs, HOG (CVPR 2005).